

Automated Skill Extraction And Resume Matching Using BERT Based NLP Model

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Abstract—In the present hiring environment, manually screening resumes and matching candidate qualifications with job criteria is time-consuming and error-prone. In order to achieve two primary objectives, this study provides a comprehensive system that employs a refined BERT-based model: (1) automatically extracting skills from job descriptions and resumes, and (2) thoroughly analysing data on skills trends across multiple dimensions, such as years of experience (YOE), job position, salary, and country. Additionally, advanced text similarity metrics and adversarial training techniques are being researched to improve assessment and resilience. Our data analytics module displays statistically significant market trends, and experimental results demonstrate that the BERT-based model achieves excellent accuracy in semantic matching and named entity recognition (NER). The combination of adversarial methods and modern metrics, such as BERTScore, substantially enhances system reliability and opens up new research directions.

Keywords – Machine Learning, Natural Language Processing (NLP), BERT, keyword extraction.

I. INTRODUCTION

Conventional recruitment methods have historically relied on human resume evaluation and rule-based keyword matching. these approaches frequently neglect the subtleties of natural language, resulting in inadequate candidate selection. recent advancements in deep learning, particularly the emergence of models based on transformers such as BERT (bidirectional encoders) reflect have transformed natural language processing (NLP) by offering sophisticated, context-sensitive embeddings.

the increase in internet job listings has produced extensive unstructured data, the examination of which can provide insights into regional, industry, and experience trends in skill demands. this research seeks to automate the processes of skill extraction and matching, as well as the following analysis of market trends.

skill extraction and matching automate the identification of skill-related items from job descriptions and resumes utilizing a fine-tuned BERT model. develop a semantic matching algorithm that measures the alignment between candidate skills and job specifications.

consolidate and examine job posting data to comprehend the demand for talents across various geographies, salary brackets, roles, and years of experience (YOE).

It incorporates cutting-edge text similarity metrics (e.g., BERTScore) and adversarial training methodologies to assess and enhance model robustness and dependability.

II. LITERATURE REVIEW

Akkasi et al. [1] initial methods in resume screening predominantly depended on rule-based algorithms and static keyword matching utilising established skill dictionaries, which encountered challenges related to synonymy and contextual interpretation. The integration of contextual information by utilization of techniques like Conditional Random Fields (CRFs) and Support Vector Machines (SVMs) represented a significant improvement, although required extensive feature engineering. Subsequently, advances such as word embeddings and Long Short-Term Memory networks (LSTMs) enhanced the representation of semantic links; but, their unidirectional characteristics imposed constraints. The advancement of transformer-based models has resulted in significant enhancements, facilitating the extraction of semantically rich representations for skills. Recent research indicates that fine-tuning models like BERT on domain-specific data markedly improves performance in Named Entity Recognition (NER) tasks.

Sun et al. [2] Recent research has broadened the evaluation of text similarity by including metrics that account for both syntactic and semantic distinctions. Initial methodologies, such as TER (Translation Edit Rate) and ITER, assess proximity by quantifying the edits necessary to convert a candidate text into a reference. Alternative approaches such as PER and CDER consider block permutations in text. Contemporary methodologies utilise dense word embeddings (MEANT, YISI-1) to encapsulate semantic similarity.

A notable advancement in this domain is BERTScore, which utilises contextual embeddings from BERT to calculate cosine similarity between tokens from candidate

and reference texts. Incorporating inverse document frequency (IDF) weights, BERTScore demonstrates superior correlation with human evaluations compared to conventional metrics, albeit at higher computational expenses.

Wang et al. [3] Machine learning models, such as those utilising BERT, are susceptible to adversarial examples—inputs that seem normal to humans yet result in misclassification by the algorithms. Algorithms like TEXTFOOLER have been developed to create adversarial instances by pinpointing essential words (utilising gradient-based techniques) and substituting them with appropriate synonyms while preserving the overall meaning. Training with adversarial examples has demonstrated enhancement of model robustness by presenting the model with a broader array of language patterns.

Heffernan et al. [4] in their study "Unsupervised Topic Segmentation of Meetings with BERT Embeddings" tackle the issue of segmenting meeting transcripts into coherent subjects in the absence of labelled data. Utilizing BERT embeddings and an innovative semantic similarity scoring method, the authors attain a 15.5% decrease in error compared to unsupervised baselines and a 26.6% enhancement over supervised models trained on non-meeting text. The research underscores the constraints of supervised methods owing to the paucity of annotated meeting data and the inherent noise in transcripts. Future endeavours intend to integrate other signals such as speaker identification and meeting agendas to further improve segmentation and summarizing tasks.

Sun et al. [5] Text categorization, a core subject in Natural Language Processing (NLP), has been extensively examined using several neural networks. Early approaches employed convolutional neural networks (CNNs) and recurrent neural networks (RNNs) for modelling text sequences. Attention techniques improved performance by proficiently capturing long-range dependencies. A significant advancement in NLP emerged with the advent of pre-trained word embeddings, such as word2vec and GloVe, which provided deep semantic representations. Contextual embeddings like CoVe and enhanced representation learning by capturing context-sensitive semantics.

The evolution of pre-trained language models revolutionized NLP activities. ULMFiT developed a transfer learning framework that achieved notable results in text classification. OpenAI GPT employed transformer architectures to generate contextual representations. However, the most substantial advancement transpired with BERT, a bidirectional transformer model pre-trained by masked language modelling and next-sentence prediction.

BERT's deep bidirectional design enabled the achievement of superior results across multiple NLP benchmarks.

Despite its efficacy, further research is required to improve BERT's fine-tuning techniques for specific tasks, such as text classification.[5]

Bilal et al. [6] Investigations on forecasting the use of online reviews have examined diverse machine learning methodologies and feature-based frameworks. Conventional

methodologies, including Linear Regression, Random Forest, and Support Vector Machines, depend significantly on manually constructed variables such as review length, rating, readability, and polarity. Nonetheless, these methodologies exhibit constrained generalization capacities.

Recent breakthroughs in deep learning, such as BERT, have demonstrated promising outcomes by obviating the necessity for manually created features. Research conducted, investigated BERT's efficacy in assessing review helpfulness, although yielded conflicting findings, underscoring deficiencies in dataset selection and sequence length optimization. This study examines these deficiencies by assessing fine-tuned BERT models with differing sequence lengths utilizing Yelp review datasets.

Thomas et al. [7] Web scraping is a method for obtaining data from webpages, sometimes known as "screen scraping" or "web data extraction." It is extensively utilized in research and business for data mining, competitive analysis, and trend monitoring.

HTML or XHTML pages frequently include unstructured data, necessitating automated extraction technologies. Scrapy, an open-source web crawling framework in Python, is frequently utilized for the collection and analysis of extensive datasets. The extensive libraries and community support of Python render it a favoured language for data-centric applications. Prominent platforms such as Amazon AWS and Google offer web scraping capabilities for effective data acquisition. Prior research underscores the significance of web structure analysis and automated query processing for efficient data extraction.

Lotfi et al. [8] Web scraping, often referred to as web extraction or web harvesting, is an essential method for gathering data from internet sources. It is extensively utilized for corporate intelligence, marketing analysis, and research across several sectors.

Recent studies emphasize the significance of sophisticated online scraping methodologies and instruments for efficient data acquisition. Scrapy distinguishes itself among these tools due to its speed, adaptability, and asynchronous request management, facilitating swift data extraction. The integrated support for CSS and XPath selectors facilitates data extraction from web pages.

Moreover, services such as ScraperAPI augment scraping efficiency by administering proxies and CAPTCHAs.

Research highlights the significance of online scraping in extracting insights from extensive web databases, facilitating informed decision-making in data-driven sectors.

Xie et al. [9] Conventional economic models, exemplified by the Cobb-Douglas production function (Cobb & Douglas, 1928), have historically functioned as essential instruments for examining business behaviour by correlating inputs and outputs. Nevertheless, these models frequently neglect the internal decision-making mechanisms within organizations, particularly in light of contemporary technological progress. Recent studies underscore the revolutionary impact of AI algorithms, including neural networks and massive language models, in enhancing operations and decision-making. The Skill-Task Matching Model enhances the conventional task-based methodology by integrating multidimensional and

iterative learning processes facilitated by artificial intelligence. This approach utilizes vectorized representations of activities, abilities, and business strategies, highlighting iterative learning using a neural network to reconcile the disparity between anticipated and actual profitability.

Chen et al. [10] The research on automated skill extraction underscores the increasing significance of Natural Language Processing (NLP) and deep learning models in recruitment and labour market analysis. Recent research highlight the efficacy of BERT-based models in extracting skills from unstructured text, including resumes and job descriptions, owing to their capacity to comprehend contextual links in language. The Deep BERT-based Semantic Skill Matching (DBSSM) classifier incorporates classification algorithms, domain-specific modifications, and the ESCO taxonomy to enhance precision. This model attained a MAP@1 score of 0.413 and an AF1 score of 0.206, above conventional approaches.

These innovations improve candidate shortlisting processes and provide vital insights for job seekers, educators, employers, and policymakers.

III. METHODOLOGY

The foundation of all BERT models comes from the well know 2017 paper from Vaswani et al. [11] ‘Attention is all you need’. The idea is to update word/token embeddings based on the surrounding text(token). We then process input sequences using an encoder-decoder structure with the help of self-attention blocks.

A. Data Collection

- Data was aggregated primarily from two sources:
Kaggle Datasets: Publicly available datasets provided a diverse array of job descriptions.
<https://www.kaggle.com/datasets/asaniczka/1-3m-linkedin-jobs-and-skills-2024>
- **LinkedIn Scraper:** A custom-built scraper was used to collect additional job descriptions directly from LinkedIn, ensuring a broader representation of industries and skill sets.

B. Data Labelling

Because of a lack of resources and a small team, manual annotation was not possible. Hence, we chose to use a rule-based NLP model to make BIO labels for the collected data automatically. This method adds noise, but it worked well to make a big enough training dataset without having to pay a lot of money for hand-labelling.

C. Model Architecture:

Input Representation

Each sentence is first segmented into words using a custom tokenization strategy. By preserving whole words as tokens, we ensure a one-to-one mapping between words and their corresponding BIO labels. The final input representation is formed by summing three components:

- **Word Embeddings:** Each token (word) is mapped to a dense vector that captures its semantic meaning.
- **Positional Embeddings:** These embeddings encode the order of tokens in the sentence.
- **Segment Embeddings:** For single-sequence tasks such as NER, segment embeddings are uniformly applied.

This approach simplifies the labelling process since each word retains its original boundaries without the need to propagate labels across multiple subword units.

The system comprises three primary modules:

1. Employment Data Extraction Module
2. BERT-Driven Processing Pipeline
3. Trend Analysis Module

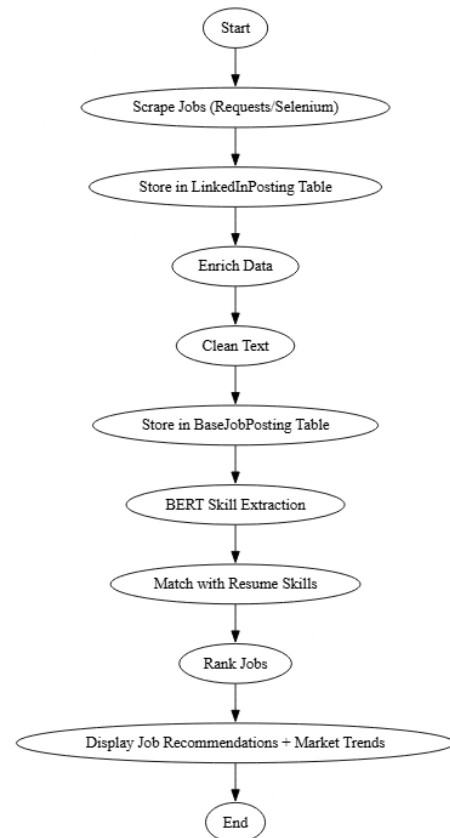


Figure 3.1: Flowchart of Application processes

D. Transformer Encoder Layers

At the core of our model is the BERT architecture, consisting of 12 transformer encoder layers. Each layer comprises two primary sub-components:

- **Multi-Head-Self-Attention:** This mechanism enables the model to learn contextual relationships by computing attention scores across all tokens in the sentence. Given our whole-word tokenization, attention is computed directly over complete words, allowing the model to capture broader contextual dependencies without the noise introduced by fragmented sub word tokens.

- to a message queue instance (RabbitMQ) also in the cloud

B. Notification Service

The notification service is a simple python script to read from the message queue and send it to the frontend GUI for a live job feed. The notification service might look unnecessary but is required because frontend (browsers) do not support MQTT protocol which is used by RabbitMQ.

Timestamp	Job Title	Company	Status	Location	Apply Link
1739841120.0243...	Senior SDE - Platform	Pine Labs		India	Apply Here
1739841118.5668...	Software Engineer	Microsoft Power Platform		India	Apply Here
1739840149.967...	Frontend Developer	Jain Software		India	Apply Here
1739840148.597...	React.js Developer - Frontend Technolo...	GAMMASTACK		India	Apply Here
1739840147.2525...	Frontend Developer	Grappus		India	Apply Here
1739840145.86555	Front End Developer	Meyrakee Advisors		India	Apply Here
1739840138.045...	Frontend Developer	ZODP		India	Apply Here
1739840136.660...	Web Developer	Web Solution Centre		India	Apply Here
1739840135.278...	Web Developer	CapriTech Global Servic...		India	Apply Here
1739840133.919...	HTML Developer	Digitalantics-pbltrcm		India	Apply Here
1739840126.0781...	Frontend developer	BlitzenX - Rapid Consult...		India	Apply Here
1739840124.677...	Web Developer	Tech India Infotech		India	Apply Here
1739840123.3371...	Web Developer	Decoriane		India	Apply Here
1739840121.9757...	PHP Developer	Amritsar Digital Academy		India	Apply Here
1739840117.7216...	Frontend Developer	Han River Technology Pv...		India	Apply Here
1739840110.239...	Web Developer	ComX Softtech Pvt Ltd		India	Apply Here
1739840090.608...	Web Application Engineers	SFO Technologies		India	Apply Here
1739840083.012...	Web Developer	21K School		India	Apply Here

Figure 4.2 Live Job Portal

C. Consumer Service

The consumer service picks datapoints from the Postgres instance and extracts job descriptions. These descriptions are then moved to the token-classification queue to extract skills using the BERT model. Note that these datapoints are only used for inference and not further fine tuning. Once done the consumer service adds this information to the graph database.

The graph DB has relations defined within skills, jobs, countries, experience level etc. It gets built slowly over time and can be visualised like a map where similar things connect with edges and dissimilar do not.

D. Analysis and Visualization

The visualization of the graph can be done easily using prebuilt modules provided by Neo4J and many other graph libraries.

Keeping the entries arranged in such a manner also enables us to answer queries like:

- What's the most asked skill in a role?
- What skills are being asked by domain?
- How many jobs are being posted on average?
- Which are the top recruiting companies?
- **What are top 10 jobs for a given set of skills?** This enables resume matching. We simply need to extract skills from resume and query our GraphDB.

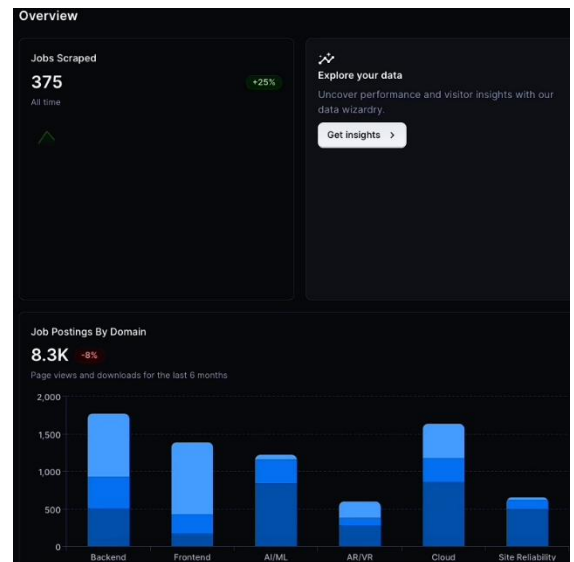


Figure 4.3 Live Trends Visualization

Result: Exhibit real-time employment suggestions and market dynamics on the frontend.

V. RESULT AND FUTURE SCOPE OF WORK

- **Model Training:** The BERT model was fine-tuned on 1.4 million LinkedIn job postings using PyTorch.

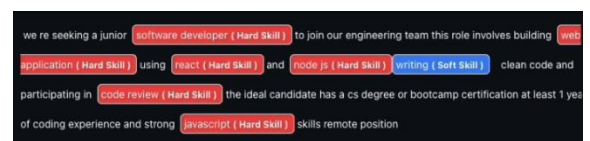


Figure 5.1 Token Classification using BERT

- **Data Pipeline:** Scraped data was pre-processed using NLTK and stored in PostgreSQL. The consumer service processed skills through the classification queue and populated a Neo4j graph database.
- **Skill Matching:** Queries to the graph database were optimized using Cypher.
- **Performance Evaluation:** Precision, Recall, and F1-score were calculated. Technical skill extraction achieved an F-score of 74%, and non-technical skill extraction reached 69%. Overall resume matching accuracy was 81%.
- **Comparison:** The proposed system outperformed traditional CRF and deep learning hybrid models.

The current system demonstrates robust performance in skill extraction and resume matching using a BERT-based NLP model and graph database integration. Future research can explore the following advancements:

- **Dataset Expansion:** Incorporating multi-domain job descriptions and resumes to enhance model generalization across industries.
- **Multilingual Support:** Fine tuning the BERT model for different languages to support a global job market.
- **Explainability Enhancement:** Implementing SHAP or LIME methods to provide clearer insights into the model's decision-making process.
- **Model Optimization:** Leveraging techniques such as knowledge distillation and parameter pruning to improve inference speed.
- **User Feedback Integration:** Utilizing feedback loops from job seekers to continuously refine matching algorithms.
- **Real-Time Recommendations:** Enhancing the graph database with streaming data pipelines for instant job recommendations.
- **Skill Gap Analysis:** Developing tools to identify and suggest relevant courses to bridge skill gaps for candidates.

These directions aim to increase the system's versatility, accuracy, and real-world applicability in dynamic job markets.

VI. CONCLUSION

The proposed system effectively integrates BERT based NLP with a graph database to enhance skill extraction and resume-job matching. The results indicate significant performance gains over conventional approaches. Future work includes expanding the datasets, optimizing the graph query engine, and integrating user feedback for continuous model improvement.

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CERTIFICATE OF PARTICIPATION

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